

Trainings ▾

General Information

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Undersize main bearings are designed with a -0.076 mm [-0.003 in] undersize profile. These bearings were developed as a cost effective repair option to extend the life of a cylinder block that exhibits main cap fretting. The undersize bearings increase the crankshaft clearance and the volume of oil flow. With the increase in oil flow, oil pressure is slightly reduced; so it is recommended that **only** one undersize bearing kit be used per engine. Bearing shells are marked with the respective part number and are offered as a kit of one upper and lower shell. Undersize bearing shells can be identified by part number or a -0.076 mm next to the part number.

Undersized bearings have been thoroughly tested with excellent results. Bearings that have been in service for a period of time will exhibit uneven copper exposure on the lower shell. This is due to a slight block bore change in geometry during fretting, with the increase oil volume of flow, undersized bearings are lubricated and cooled at a higher rate. Undersize main bearings that have been in service for a period of time will exhibit an uneven wear pattern or copper exposure. This is normal wear.

Note : Plastigage™ procedures will not accurately measure bearing oil film clearance in undersize main bearings.

Use the following procedure for more information on main bearing cap fretting and Plastigage™ use. Refer to Procedure 001-999 in Section F.
(/qs3/pubsys2/xml/en/procedures/35/35-001-999.html)

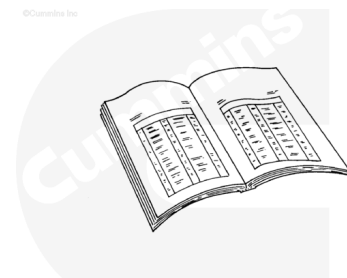
Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

Cummins Inc. recommends replacing the thrust bearings when the main bearings are replaced. Use the following procedure in the ISM, ISMe and QSM11 Service Manual, Bulletin 3666322.

Refer to Procedure 001-007 in Section 1

(/qs3/pubsys2/xml/en/procedures/35/35-001-007-tr.html)

Use the following procedure in the M11 Series Engines (STC, CELECT™, CELECT™ Plus Models) Troubleshooting and Repair Manual, Bulletin 3666139, for thrust bearing replacement.

Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/36/36-001-007.html)

- Drain the lubricating oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Remove the block stiffener plate, if equipped. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-089-tr.html)

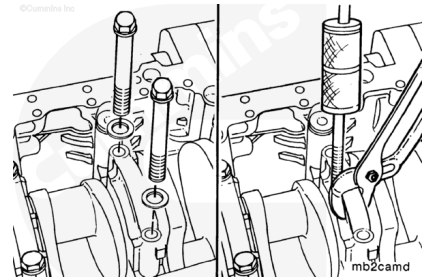
Remove

Replace the main bearings one at a time.

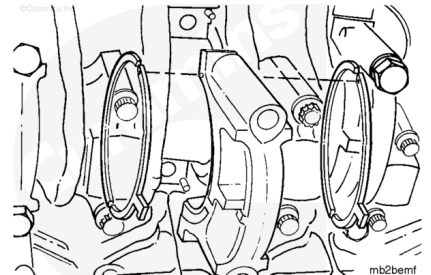
Remove the main bearing capscrews.

Use main bearing cap puller, Part Number ST-1178, to remove the cap.

The tool **must** be centered on the cap. Pull straight down to remove the cap.

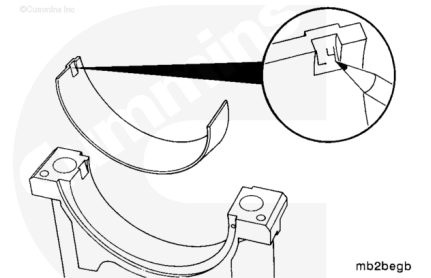


The number 4 main bearing cap has thrust bearings. The cap is located with two dowel pins.



Remove the lower main bearing shell from the cap.

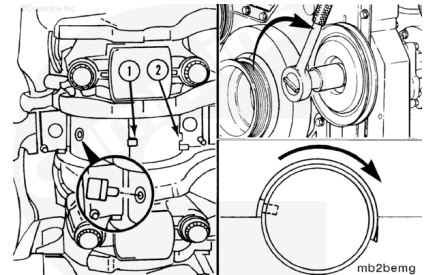
Mark the bearing shell with the letter L (lower) and the journal number from which it was removed.



To remove the upper main bearing shell, install bearing rollout tool, Part Number 3823818, in the oil hole of the main bearing journal.

Remove the tang side (1) of the main bearing shell first.

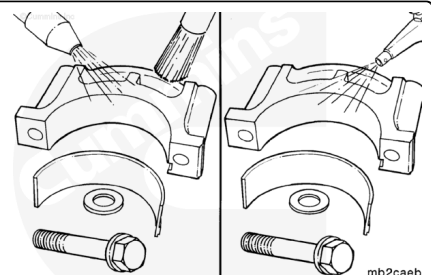
Rotate the crankshaft with the accessory drive pulley to remove the upper main bearing shell (2).



Clean and Inspect for Reuse

⚠ CAUTION ⚠

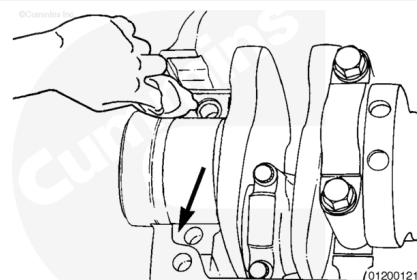
Prevent oil seepage down onto the mating faces of the block mains, cleaning of the block main capscrew threaded holes is critical. (Use cotton swabs, or equivalent, to absorb as much oil as possible prior to installing main caps.) Loctite™ 518 anaerobic sealant will not seal to an oily surface.



Note : Loctite® 518 anaerobic sealant is available at automotive parts stores. There is not a Cummins® part number available.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean all main bearing cap mounting surfaces on the cylinder block with a solvent-based degreaser.

Dry the main bearing cap mounting surfaces on the cylinder block with a clean, lint-free cloth.

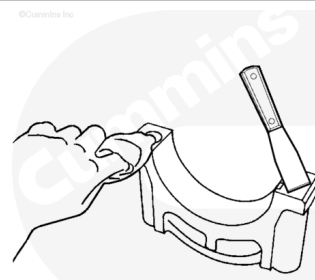
The main bearing cap surfaces **must** be free of grease and oily residue.

Use solvent and a soft bristle brush to clean the bearing cap, capscrews, washers, and bearings.

Dry the parts with compressed air.

Clean the mounting surfaces of the main bearing caps with a solvent-based degreaser. Dry the mounting surfaces of the main bearing caps with a clean, lint-free cloth.

The mounting surfaces of the main bearing caps **must** be free of grease and oily residue.



Inspect the main cap, capscrews, and washers for damage.

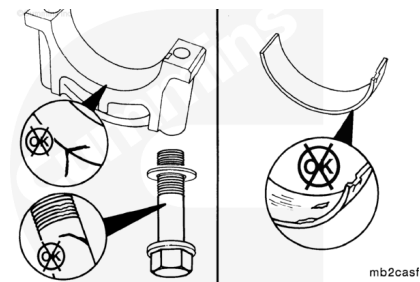
Inspect the bearing shells for nicks, scratches, or other damage.

If the main bearings are damaged, inspect the crankshaft. If the crankshaft is damaged, the engine **must** be removed for repair. Refer to Procedure 000-001 in Section 0.

(/qs3/pubsys2/xml/en/procedures/35/35-000-001-tr.html)

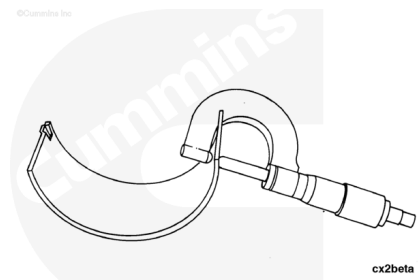
Use the following procedure in the M11 Series Engines (STC, CELECT™, CELECT™ Plus Models) Base Engine Troubleshooting and Repair Manual, Bulletin 3666139. Refer to Procedure 000-001 in Section 0.

(/qs3/pubsys2/xml/en/procedures/36/36-000-001.html)



Use an outside diameter ball-tipped micrometer to measure the main bearing shell thickness.

Standard Main Bearing Shell Thickness		
mm		in
3.894	MIN	0.1533
3.945	MAX	0.1553

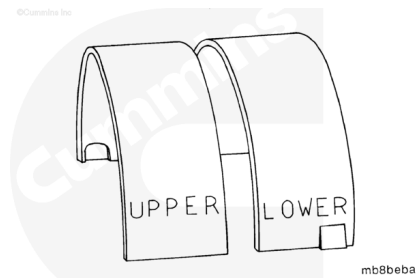


For more detailed information on bearing damage, consult the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387.

Install

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the upper and lower bearings must be installed in the correct location. The upper bearing has an oil groove. The bearing shells are marked with the words "upper" and "lower" for identification.



⚠ CAUTION ⚠

To correctly position the bearing and prevent engine damage, the bearing tang (1) must be in the slot (2) of the bearing saddle.

⚠ CAUTION ⚠

Only use Loctite™ 518 anaerobic sealant on the main cap joint mating surface. Other sealants can become hard, brittle, and allow oil and debris into the main bearing/block joint.

⚠ CAUTION ⚠

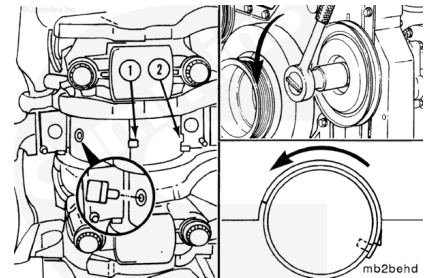
The bead must be 3 to 5 mm [0.12 to 0.2 in] wide and must not enter the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.

Used bearings **must** be installed in the same location from which they were removed.

Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the inside diameter of the upper main bearing shell.

Install the upper main bearing shells (2) using the same method used to remove them.

The tang side (1) of the bearing **must** go into place last.



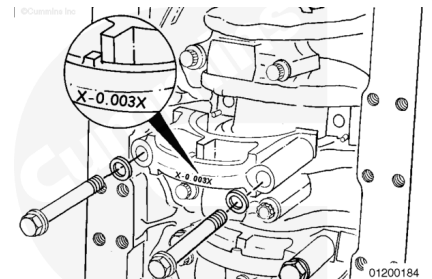
⚠ CAUTION ⚠

Low oil pressure faults can occur if more than one set undersize main bearings are installed in the same engine.

Note : Both upper and lower undersize main bearing shells **must** be installed as a set in the same block location.

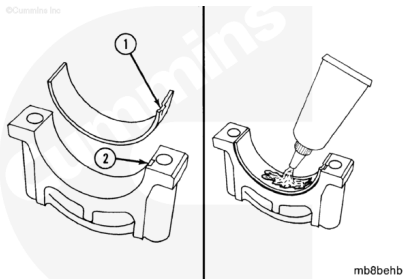
Undersize main bearing installation procedures are the same as standard main bearings procedures. Both upper and lower undersized main bearing shells **must** be installed as a set in the same block location.

For undersize main bearing installation, mark the outside back of the main cap X -0.003 X with a permanent white Dykem™ marker to identify bearing location for future repairs. Allow the Dykem™ to dry before oil is added.



Install the lower main bearing shells with the bearing tang (1) in the slot (2) of the main bearing cap.

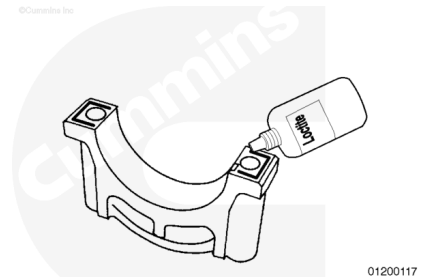
Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the inside diameter of the bearing shells.



Apply Loctite® 518 anaerobic sealant to four locations on each main bearing cap.

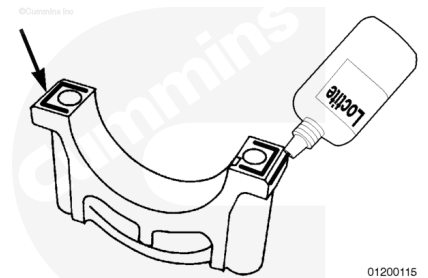
Apply a bead of Loctite® 518 anaerobic sealant to the main bearing cap mounting surface, as shown.

Repeat this procedure on the opposite side of the main bearing cap.



⚠ CAUTION ⚠

The bead must be 3 to 5 mm [0.12 to 0.2 in] wide and must not enter the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.

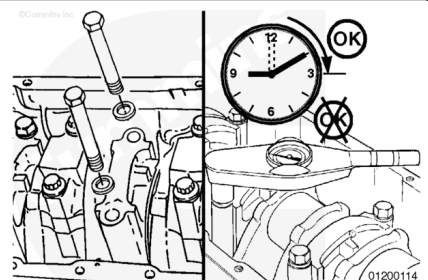


Locate the leading edge of the main bearing cap clearance chamfer.

Apply a bead of Loctite® 518 anaerobic sealant to the leading edge of the main bearing cap clearance chamfer (as shown).

Repeat this procedure on the opposite side of the main bearing cap.

To reduce the possibility of bore size issues, clearance issues, or both, the main bearing capscrews **must** be tightened within 15 minutes of the Loctite® 518 anaerobic sealant application.

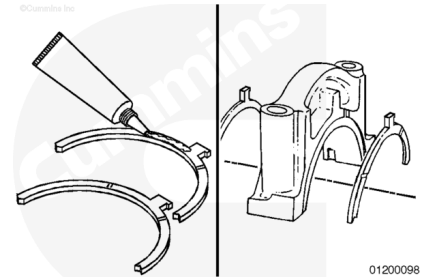


Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the lower thrust bearings.

Install the lower thrust bearings in the number 4 main bearing cap, as shown.

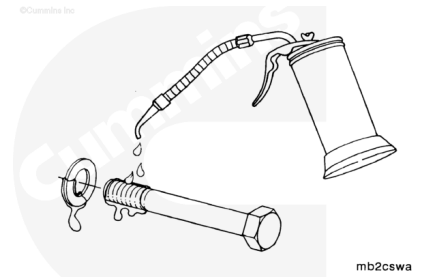
The grooves of the thrust bearing **must** be toward the crankshaft.

The locating dowels **must not** protrude above the thrust bearing surface.

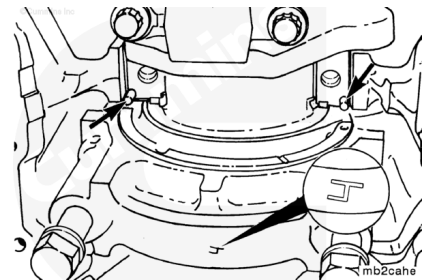


Use clean 15W-40 oil to coat the capscrew threads and both sides of the washers.

Drain the excess oil from the capscrews before installing them in the cylinder block.

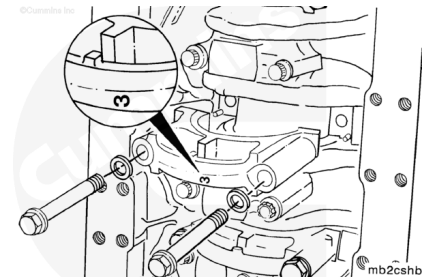


The number 4 main bearing cap **must** be aligned with the dowel pins in the bearing saddle when the capscrews are tightened.

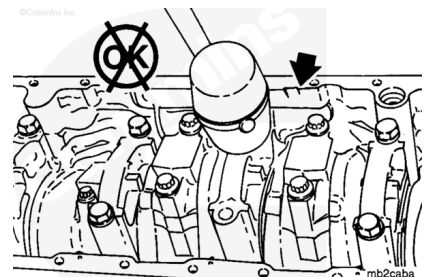


The main bearing caps are numbered 1 through 7 from front to rear in the cylinder block. The caps **must** be installed so the number on the cap matches the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap **must** be on the same side.

Install the main bearing caps.

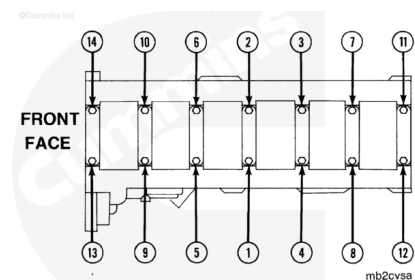


Do **not** hit the main bearing cap with a hammer.



⚠ CAUTION ⚠

When using new main bearing caps with older M11 and ISM cylinder blocks, only use the torque-plus-angle method when tightening the main bearing capscrews. The torque-to-yield procedure will damage cylinder blocks stamped 00622080 and lower.



Torque Plus Angle procedure (Engine Block Numbers Lower Than 00622081)

Tighten the main bearing capscrews in alternating sequence to the following torque values:

Torque Value:

1. 68 n•m [50 ft-lb]
2. 142 n•m [105 ft-lb]
3. 210 n•m [155 ft-lb]
4. Loosen completely
5. Repeat steps 1 through 3.

Torque-to-Yield Procedure

Note : Only use the torque-to-yield procedure to tighten the main bearing cap capscrews on the following engines:

- Engine serial numbers 35011095 and higher.
- Engine cylinder blocks stamped 00622081 and higher.
- Blocks with TTY stamped on the rear engine serial number stamp pad.

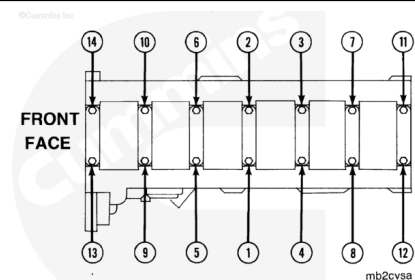
Tighten the main bearing cap capscrews with the following steps:

Torque Value:

1. 95 n•m [70 ft-lb]
2. 135 n•m [100 ft-lb]
3. Loosen completely
4. 95 n•m [70 ft-lb]
5. 135 n•m [100 ft-lb]
6. Rotate 180 degrees.

To rotate 180 degrees, use a socket that has been etched with two axial lines 180 degrees apart.

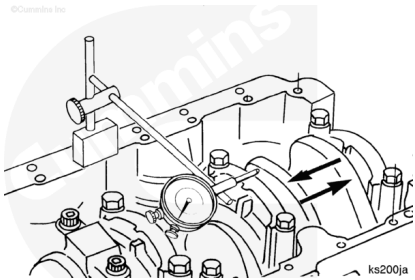
Place the socket over the main cap capscrew head, and mark the cap with a lead pencil at the location of the socket axial line number 1.



Use an impact wrench to rotate the main bearing capscrew in the direction of tightening until the axial line number 2 lines up with the mark on the cap (180 degrees).

Use a dial indicator to measure the crankshaft end clearance.

Crankshaft End Clearance		
mm		in
0.10	MIN	0.004
0.56	MAX	0.022



If the end clearance is **not** within the specifications, inspect the main bearing cap and thrust bearing surfaces in the cylinder block.

Finishing Steps

⚠ CAUTION ⚠

The engine must have adequate oil pressure within 15 seconds after starting. If the warning light indicating low oil pressure has not gone out or there is no oil pressure indicated on a gauge within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level is in the oil pan.



- Install the block stiffening plate, if equipped. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-089-tr.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Operate the engine to normal operating temperature and check for leaks.

